

Online Supplement to “Market Integration, Platformization, and Local Multipliers”

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A Proofs and Demand Details

Proof of Lemma 1. Write $\gamma = 1 - \alpha - \beta > 0$. The first-order conditions for the outer household problem are

$$\frac{\alpha}{M} = \lambda P_M, \quad \frac{\beta}{N} = \lambda, \quad \frac{\gamma}{Z} = \lambda.$$

The budget constraint gives $\lambda = 1/Y$, hence expenditures on (M, N, Z) equal $(\alpha Y, \beta Y, \gamma Y)$. Substitution into utility gives

$$R = K \frac{Y}{P_M^\alpha}, \quad K = \alpha^\alpha \beta^\beta \gamma^\gamma. \quad (\text{A.1})$$

Let $h_P(p_A, p_P, 1)$ be the platform Hicksian demand for one unit of the homothetic retail aggregate. Differentiability of the unit-expenditure function and Shephard’s lemma imply

$$\frac{\partial P_M}{\partial p_P} = h_P(p_A, p_P, 1), \quad s = \frac{p_P h_P(p_A, p_P, 1)}{P_M} = \frac{\partial \ln P_M}{\partial \ln p_P}.$$

Because p_A is fixed and $d \ln p_P / dz = -1$, the chain rule yields $d \ln P_M / dz = -s$, proving the demand identities in the lemma. $\square$

Proof of Lemma 2. For fixed (z, κ, G) , define

$$T(Y) = B + \omega G + (\beta + \alpha \omega)Y.$$

Since $s \in (0, 1)$ and both retention shares lie in $[0, 1]$, $0 \leq \omega \leq 1$. Thus T maps $\mathbb{R}_+$ into itself and

$$|T(Y) - T(\tilde{Y})| = (\beta + \alpha \omega)|Y - \tilde{Y}|, \quad \beta + \alpha \omega \leq \alpha + \beta < 1.$$

The contraction theorem gives a unique fixed point and convergence of iteration from every non-negative initial value. Solving the fixed-point equation yields (7). Its numerator is positive, while $D \geq 1 - \alpha - \beta > 0$, so $Y > 0$. Lemma 1 then determines the remaining household objects uniquely. $\square$

A.1 Benchmark multipliers

Holding (z, κ, G) fixed and varying the realized autonomous-income injection or outside-funded retail spending gives

$$\frac{\partial Y}{\partial B} = \frac{1}{D}, \quad \frac{\partial Y}{\partial G} = \frac{\omega}{D} = \omega \sum_{n=0}^{\infty} (\beta + \alpha \omega)^n.$$

The series converges because $\beta + \alpha \omega < 1$. It records that retail leakage recurs in every induced spending round.

Proof of Proposition 1. The ratio identities follow immediately from $Y = B/D$ and $R = KY/P_M^\alpha$. Organizational capacity does not affect B_j , s_j , or P_{Mj} . Differentiating

$$\mathcal{D}(\kappa) = \frac{D_1(\kappa)}{D_0(\kappa)}$$

therefore gives

$$\begin{aligned} \frac{\partial \ln \mathcal{D}(\kappa)}{\partial \kappa} &= -\alpha \delta_\rho \left[\frac{s_1}{D_1(\kappa)} - \frac{s_0}{D_0(\kappa)} \right] \\ &= -\frac{\alpha \delta_\rho d_A (s_1 - s_0)}{D_1(\kappa) D_0(\kappa)} < 0. \end{aligned} \quad (\text{A.2})$$

Hence both finite income ratios are strictly increasing in κ .

For any $q \in (\mathcal{D}(1), \mathcal{D}(0))$, continuity and strict monotonicity give a unique $\kappa(q) \in (0, 1)$ satisfying $\mathcal{D}(\kappa(q)) = q$. Solving

$$d_A + \alpha s_1 \Delta(\kappa) = q[d_A + \alpha s_0 \Delta(\kappa)]$$

gives (12). The denominator in that expression is positive at a solution because

$$s_1 - q s_0 = \frac{s_1 D_0(\kappa) - s_0 D_1(\kappa)}{D_0(\kappa)} = \frac{d_A (s_1 - s_0)}{D_0(\kappa)} > 0.$$

Since $0 < \mathcal{P} < 1$, $q_R^F = \mathcal{B}/\mathcal{P}^\alpha > q_Y^F = \mathcal{B}$. The inverse of the strictly decreasing function $\mathcal{D}$ is decreasing, so $\kappa_R^F < \kappa_Y^F$. Comparing $\mathcal{D}(\kappa)$ with the two targets proves the three strict regions. At the first threshold,

$$\frac{R_1}{R_0} = 1, \quad \frac{Y_1}{Y_0} = \mathcal{P}^\alpha < 1;$$

at the second,

$$\frac{Y_1}{Y_0} = 1, \quad \frac{R_1}{R_0} = \mathcal{P}^{-\alpha} > 1.$$

□

Proof of Corollary 1. At $G = 0$,

$$Y = \frac{B}{D}, \quad D = d_A + \alpha s \Delta(\kappa), \quad D_\kappa = -\alpha s \delta_\rho.$$

Therefore $\partial \ln Y / \partial \kappa = \alpha s \delta_\rho / D$. Since $D_z = \alpha \Delta(\kappa) s_z$,

$$\frac{\partial^2 \ln Y}{\partial z \partial \kappa} = \alpha \delta_\rho \frac{s_z D - s D_z}{D^2} \quad (\text{A.3})$$

$$= \frac{\alpha \delta_\rho d_A s_z}{D^2} > 0. \quad (\text{A.4})$$

Finally, $\ln R = \ln K + \ln Y - \alpha \ln P_M$, and P_M is independent of κ , so the same cross-partial holds for $\ln R$. □

Proof of Proposition 2. Strict concavity, the endpoint restrictions, and interior positivity imply $g'(0) > 0$ and $g'(1) < 0$. For example, for every $u \in (0, 1)$,

$$g'(0) \geq \frac{g(u) - g(0)}{u} > 0, \quad g'(1) \leq \frac{g(1) - g(u)}{1 - u} < 0.$$

Write $b = \alpha\Delta(\kappa) > 0$ and define

$$F(s, b) = g'(s)(d_A + bs) - bg(s).$$

Direct differentiation gives

$$\frac{\partial \Lambda_g}{\partial s} = \frac{bF(s, b)}{(d_A + bs)^2}, \quad F_s(s, b) = g''(s)(d_A + bs) < 0. \quad (\text{A.5})$$

Moreover,

$$F(0, b) = d_A g'(0) > 0, \quad F(1, b) = (d_A + b)g'(1) < 0.$$

Thus F has exactly one zero $s^*(b) \in (0, 1)$. Equation (A.5) shows that Λ_g rises before this zero and falls after it. Since its endpoint values are zero and it is positive in the interior, s^* is the unique global maximizer.

If $g(s) = g(1 - s)$, differentiability gives $g'(1/2) = 0$, so $F(1/2, b) = -bg(1/2) < 0$. Because $F(0, b) > 0$ and F is strictly decreasing, $s^*(b) < 1/2$.

For the capacity comparative statics, evaluate expressions at $s = s^*(b)$ and write $D^* = d_A + bs^*$. The first-order condition implies $g'(s^*) = bg(s^*)/D^*$, and hence

$$F_b(s^*, b) = s^* g'(s^*) - g(s^*) = -\frac{d_A g(s^*)}{D^*} < 0.$$

The implicit-function theorem, $b_\kappa = -\alpha\delta_\rho$, and the envelope theorem yield

$$\frac{\partial s^*}{\partial \kappa} = -\frac{\alpha\delta_\rho d_A g(s^*)}{g''(s^*)[D^*]^2} > 0, \quad (\text{A.6})$$

$$\frac{\partial \Lambda_g^{\max}}{\partial \kappa} = -\frac{\alpha\delta_\rho d_A g(s^*)}{[D^*]^2} < 0. \quad (\text{A.7})$$

□

A.2 Admissibility of the general share-speed

The share-speed restriction is consistent with a two-channel homogeneous demand system. Let $t = \ln(p_P/p_A)$ and take any interior solution of $ds/dt = -g(s)$. Define

$$\phi(r) = \exp \left\{ \int_{r_0}^r s(\ln u) d \ln u \right\}, \quad P_M(p_A, p_P) = p_A \phi(p_P/p_A).$$

Then P_M is linearly homogeneous and its platform expenditure share is $p_P P_{M, p_P} / P_M = s$. It is increasing because the two expenditure shares are s and $1 - s$. Moreover,

$$\phi''(r) = -\frac{\phi(r)}{r^2} [g(s) + s(1 - s)] < 0,$$

so P_M is a valid concave unit-expenditure function. This construction also shows that the non-CES share-speed in Figure 1 is not merely an arbitrary curve detached from a demand system.

A.3 CES specialization

The main assumptions nest the two-channel CES aggregator

$$M = \left[(1-a)^{1/\eta} M_A^{(\eta-1)/\eta} + a^{1/\eta} M_P^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}, \quad a \in (0,1), \quad \eta > 1. \quad (\text{A.8})$$

Cost minimization gives

$$P_M(z) = \left[(1-a)p_A^{1-\eta} + ap_P(z)^{1-\eta} \right]^{1/(1-\eta)}, \quad (\text{A.9})$$

$$s(z) = \frac{ap_P(z)^{1-\eta}}{(1-a)p_A^{1-\eta} + ap_P(z)^{1-\eta}}, \quad (\text{A.10})$$

$$s'(z) = (\eta-1)s(1-s). \quad (\text{A.11})$$

Thus $g(s) = (\eta-1)s(1-s)$ satisfies all conditions in Proposition 2, including symmetry. With $t(\kappa) = \sqrt{1 + \alpha\Delta(\kappa)/d_A}$, its unique peak and height have the closed forms

$$s^*(\kappa) = \frac{1}{1+t(\kappa)} < \frac{1}{2}, \quad (\text{A.12})$$

$$\Lambda_g^{\max}(\kappa) = (\eta-1) \frac{t(\kappa) - 1}{t(\kappa) + 1}. \quad (\text{A.13})$$

These expressions are the CES specialization, not assumptions needed for the general peak result.

As $\eta \downarrow 1$, the CES system approaches the Cobb–Douglas channel case, $P_M \rightarrow p_A^{1-a} p_P^a$ and $s \rightarrow a$. At fixed unequal prices, $\eta \rightarrow \infty$ concentrates expenditure on the cheaper channel. These are limits of the specialization rather than finite-parameter corners in the main interior demand path.

Figure S1 combines this CES specialization with the marginal-income identities in the main text. It is a local derivative map, not a finite-change calculation: for each fixed capacity level, the position of $\Lambda_g(s, \kappa)$ relative to ε and $\varepsilon + \alpha s$ determines the signs of $d \ln Y/dz$ and $d \ln R/dz$. Movement to the right can represent deeper integration only conditional on the maintained mapping $ds/dz > 0$ and fixed remaining parameters. At finite CES prices the share lies strictly in $(0, 1)$; the displayed endpoints are the continuous extension used by the peak theorem. The CES weight a and the two price levels determine the mapping from z to s , but do not enter this geometry once s is placed on the horizontal axis.

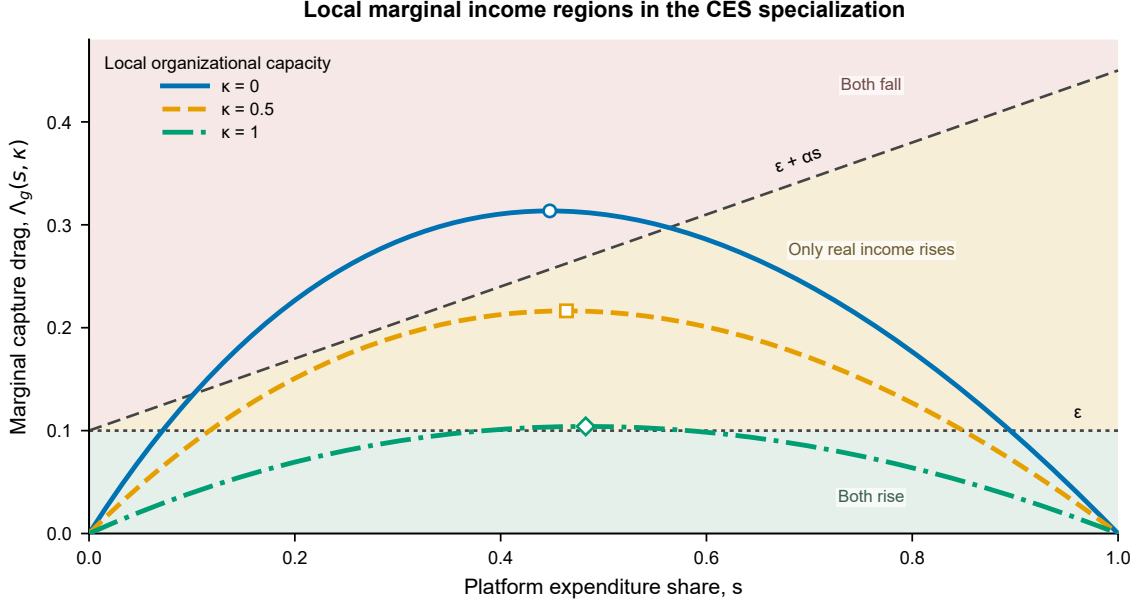


Figure S1: Local marginal income regions in the CES specialization. Under $G = 0$, the curves plot $\Lambda_g(s, \kappa)$ at $\kappa = 0, 0.5, 1$ for $g(s) = 3s(1 - s)$, $\alpha = 0.35$, $\beta = 0.25$, $\eta = 4$, $\epsilon = 0.10$, $\rho_A = 0.8$, and $(\rho_P, \bar{\rho}_P) = (0.1, 0.6)$. The horizontal line is ϵ ; the upward-sloping line is $\epsilon + \alpha s$. Below ϵ , nominal local income and consumption-equivalent real income both rise. Between the two reference lines, nominal local income falls while consumption-equivalent real income rises. Above $\epsilon + \alpha s$, both fall; equality with a reference line gives zero marginal change for the corresponding income measure. Open markers identify drag maxima, not optimal platform shares. The regions classify local marginal effects, not finite changes. Illustration, not calibration.

A.4 Boundary cases and outside-funded spending

If $\Delta(\kappa) = 0$, then $\Lambda_g \equiv 0$, so no unique drag peak exists. If $\Delta(\kappa) < 0$, integration shifts expenditure toward the higher-retention channel and the capture-drag proposition does not apply. The capacity derivatives in Proposition 2 are ordinary interior derivatives; feasible one-sided versions apply at $\kappa = 0, 1$. Without symmetry of g , the unique peak remains but need not lie below one-half.

The endpoint-crossing condition in Proposition 1 is sufficient for two interior capacity thresholds. A target strictly outside $[\mathcal{D}(1), \mathcal{D}(0)]$ has no threshold on $[0, 1]$; equality with an endpoint places the threshold at $\kappa = 1$ or 0 , respectively. Only a target in $(\mathcal{D}(1), \mathcal{D}(0))$ yields an interior threshold. The finite ratio identities remain valid in every case.

For completeness, when nominal outside-funded spending $G > 0$ is fixed,

$$\frac{d \ln Y}{dz} = \frac{\epsilon B + \omega_z G}{B + \omega G} + \frac{\alpha \omega_z}{D}, \quad (\text{A.14})$$

$$\frac{d \ln R}{dz} = \frac{\epsilon B + \omega_z G}{B + \omega G} + \frac{\alpha \omega_z}{D} + \alpha s, \quad (\text{A.15})$$

$$\frac{\partial \ln Y}{\partial \kappa} = s \delta_\rho \left[\frac{G}{B + \omega G} + \frac{\alpha}{D} \right] > 0, \quad \frac{\partial \ln R}{\partial \kappa} = \frac{\partial \ln Y}{\partial \kappa}. \quad (\text{A.16})$$

For a finite change, define

$$\mathcal{N} = \frac{B_1 + \omega_1 G}{B_0 + \omega_0 G}.$$

Then $Y_1/Y_0 = \mathcal{N}/\mathcal{D}$ and $R_1/R_0 = \mathcal{N}/(\mathcal{D}\mathcal{P}^\alpha)$. Because $\mathcal{N}$ itself varies with capacity, the closed-form M1 capacity thresholds are stated for $G = 0$.